

ORIGINAL RESEARCH

Angiographic Quantitative Flow Ratio-Guided Coronary Intervention



5-Year Follow-Up From the FAVOR III China Randomized Trial

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ABSTRACT

BACKGROUND The multicenter, randomized, sham-controlled FAVOR III China trial (Comparison of Quantitative Flow Ratio-Guided and Angiography-Guided Percutaneous Intervention in Patients with Coronary Artery Disease) demonstrated that quantitative flow ratio (QFR)-guided percutaneous coronary intervention (PCI) resulted in better outcomes compared with angiographic guidance at 1-year and 2-year follow-up. Whether these benefits are sustained over long-term follow-up remains uncertain.

OBJECTIVES The purpose of this study was to evaluate the long-term effectiveness and safety of a QFR-guided PCI strategy compared with angiography-guided PCI at 5 years.

METHODS Patients with at least 1 angiographically intermediate coronary lesion (50%-90% diameter stenosis) in a vessel ≥ 2.5 mm diameter were randomized to a QFR-guided (PCI performed only if QFR ≤ 0.80) or angiography-guided strategy. The primary endpoint was major adverse cardiac events (a composite of all-cause death, myocardial infarction, or ischemia-driven revascularization) at 1 year; 5-year outcomes data are reported herein.

RESULTS At 5 years, major adverse cardiac events composite was lower with QFR guidance than with angiography guidance (17.5% vs 21.1%; HR: 0.80; 95% CI: 0.69-0.92; $P = 0.002$), driven by fewer myocardial infarctions (5.8% vs 9.0%; HR: 0.63; 95% CI: 0.49-0.80; $P < 0.0001$) and ischemia-driven revascularizations (9.6% vs 12.0%; HR: 0.78; 95% CI: 0.64-0.95; $P = 0.02$) in the QFR-guided group. All-cause death did not differ between groups. Landmark analysis showed that the benefit of QFR guidance accrued predominantly within the first 2 years (8.5% vs 12.5%; HR: 0.66; 95% CI: 0.54-0.81; $P < 0.0001$), with similar outcomes between 2 and 5 years (10.2% vs 11.2%; HR: 0.90; 95% CI: 0.73-1.11; $P = 0.32$; P for interaction = 0.001).

CONCLUSIONS Compared with angiography guidance, QFR-guided strategy improved 5-year clinical outcomes, with benefits primarily achieved within the first 2 years. (The FAVOR III China Study; [NCT03656848](https://doi.org/10.1016/j.jacc.2026.04.037)) (JACC. 2026;88:800-812) © 2026 by the American College of Cardiology Foundation.

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Physiological assessment has become integral to the management of stable coronary artery disease (CAD), while guiding the decision-making for percutaneous coronary intervention (PCI), aiming to relieve symptoms and optimize long-term clinical outcomes.¹⁻³ Randomized trials and guidelines consistently emphasize the importance of identifying hemodynamically significant stenoses to ensure appropriate lesion selection and avoid unnecessary interventions.⁴⁻⁶ Although pressure wire-based indices as fractional flow reserve (FFR) and instantaneous wave-free ratio, are well validated, their widespread adoption in clinical practice has been limited due to procedural complexity, additional costs, and risks associated with wire manipulation and hyperemic agents.^{7,8}

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Quantitative flow ratio (QFR) was developed as a novel, angiography-derived, wire-free index that enables real-time assessment of lesion-specific ischemia and has demonstrated diagnostic accuracy comparable with FFR.^{9,10} The investigator-initiated, multicenter FAVOR III China trial (Comparison of Quantitative Flow Ratio-Guided and Angiography-Guided Percutaneous Intervention in Patients with Coronary Artery Disease; [NCT03656848](#)) was designed to evaluate the clinical outcomes of a QFR-guided strategy compared with conventional angiography-guided PCI in patients with angiographically intermediate (50%-90%) coronary stenoses.¹¹ At 1-year, QFR guidance was superior to angiography guidance in reducing the primary endpoint of major adverse cardiac events (MACE).¹² The benefits were sustained at 2 years, with a 34% relative risk reduction in MACE, mainly driven by

reductions in myocardial infarction (MI) and ischemia-driven revascularization (ID-Revasc).¹³

In the landmark FAME (Fractional Flow Reserve Versus Angiography for Multivessel Evaluation) trial, the 1-year primary endpoint favored FFR guidance; however, this advantage was no longer apparent at 5-year follow-up, likely in part because of the relatively modest sample size.^{14,15} These findings underscore the importance of evaluating the long-term durability of physiology-guided strategies beyond early outcome differences. Furthermore, whether the long-term benefits of QFR guidance are similarly attenuated over time has never been established. The 5-year results from FAVOR III China provide a valuable opportunity to address this gap, offering key insights into the durability and underlying mechanisms of physiology-based guidance in contemporary clinical practice.

METHODS

STUDY DESIGN AND PARTICIPANTS. As previously reported, FAVOR III China was an investigator-initiated, multicenter, sham-controlled, blinded, randomized trial conducted at 26 hospitals across China.¹¹⁻¹³ The study was designed by a steering committee and an international advisory board, with oversight from an independent data and safety monitoring board. The study complied with the Declaration of Helsinki and Good Clinical Practice guidelines and approval was obtained from the ethics committee of all participating centers. All participants provided written informed consent.

ABBREVIATIONS AND ACRONYMS

CAD = coronary artery disease
FFR = fractional flow reserve
ID-Revasc = ischemia-driven revascularization
MACE = major adverse cardiac events
MI = myocardial infarction
PCI = percutaneous coronary intervention
QFR = quantitative flow ratio
RVD = reference vessel diameter
SYNTAX = Synergy between Percutaneous Coronary Intervention with Taxus and Cardiac Surgery

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

Eligible participants were adults who presented with stable or unstable angina, or recent MI (≥ 72 h), in whom PCI was planned based on angiographic assessment. Key angiographic inclusion criteria required at least 1 lesion with 50% to 90% diameter stenosis in a coronary artery with a reference vessel diameter (RVD) ≥ 2.5 mm, as visually assessed by investigators. Major exclusion criteria included severe heart failure, moderate or severe chronic kidney disease (serum creatinine >150 $\mu\text{mol/L}$ or estimated glomerular filtration rate <45 mL/min/1.73 m²), allergy to contrast agents or study medications, pregnancy, life expectancy <1 year, and angiograms unsuitable for reliable QFR assessment (eg, severe vessel tortuosity, overlap, or suboptimal image quality).

RANDOMIZATION AND MASKING. Before randomization, operators were required to declare all target lesions intended for PCI based solely on angiography. This information was recorded in the electronic data capture system. Patients were then randomized 1:1 to either the QFR-guided or angiography-guided strategy. Randomization was stratified by diabetes mellitus, multivessel disease, lesion with $>90\%$ stenosis, thrombolysis in MI flow <3 , and study center, using a secure, web-based system with block sizes of 6.

Patients and research personnel were blinded to treatment allocation. To preserve masking, patients in both groups wore music-playing headphones and waited 10 minutes for either real or sham QFR computation before PCI. Blinding was maintained for 1 year after the index procedure.

TREATMENT. In the QFR-guided group, online QFR was measured for all target vessels with RVD ≥ 2.5 mm and 50% to 90% stenosis. First-generation QFR technology was used in this study that reconstructed 3-dimensional coronary anatomy and computational fluid dynamics based on 2 angiographic projections $\geq 25^\circ$ apart. The QFR analysis was performed in real time at each site (AngioPlus system, Pulse Medical). PCI was performed only if QFR was ≤ 0.80 or if the angiographic lesion stenosis severity was $>90\%$. If all lesions had QFR >0.80 and stenosis $<90\%$, PCI was deferred and the patient was treated with optimal medical therapy alone.

In the angiography-guided group, PCI was performed according to the prerandomization plan. In both groups, lesions with $>90\%$ stenosis in vessels with RVD ≥ 2.5 mm required PCI before randomization. Staged PCI within 60 days was permitted for multivessel disease. The use of drug-eluting stents, drug-coated balloons, plain balloon angioplasty, and intravascular imaging was left to operator's discretion

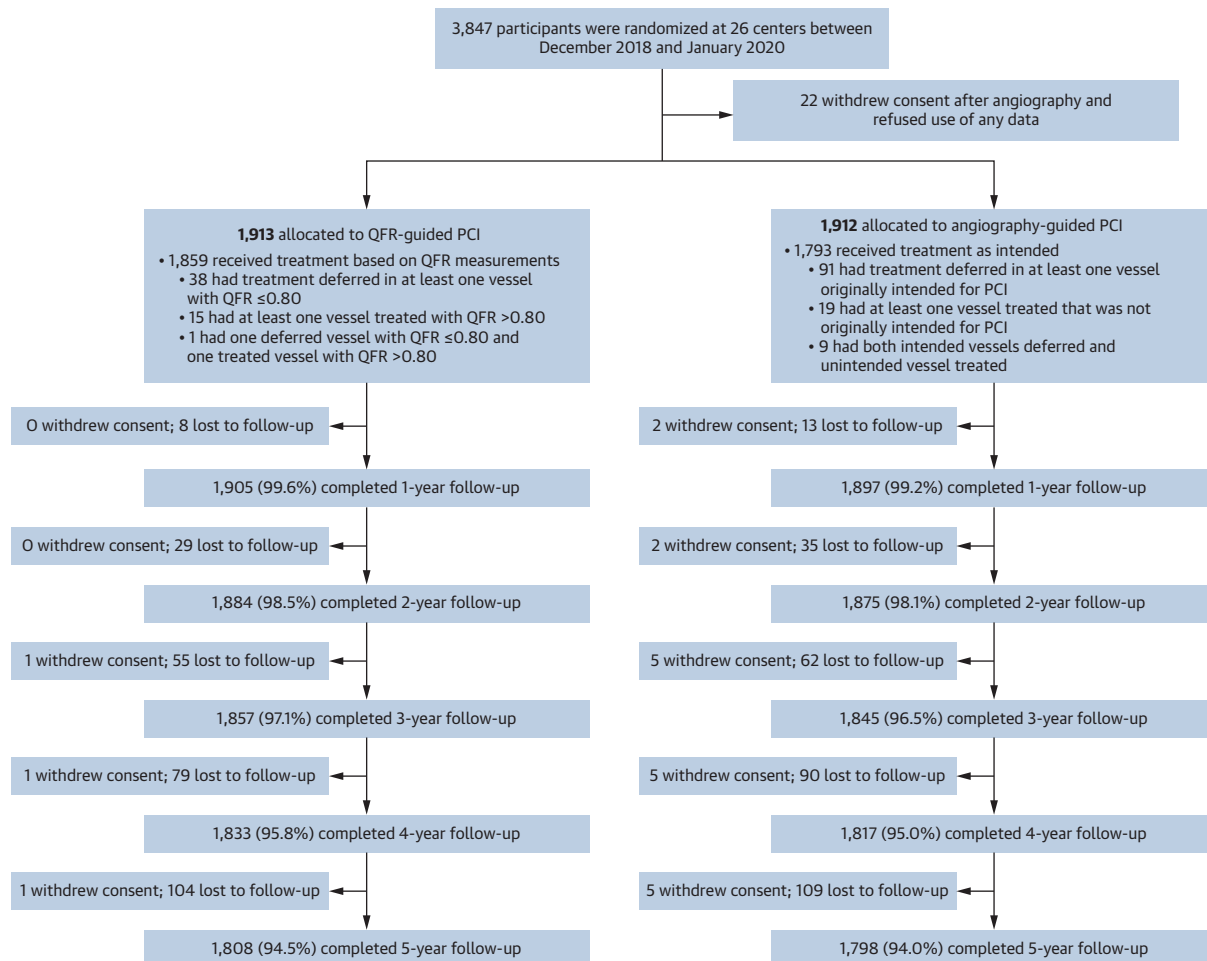
ENDPOINTS AND FOLLOW-UP. The primary endpoint of the FAVOR III China trial was the 1-year rate of MACE, defined as a composite of all-cause death, MI, or ID-Revasc. The major secondary endpoint was the 1-year rate of MACE excluding periprocedural MI arising from the index or planned staged procedures. Additional secondary endpoints included individual components of MACE, cardiovascular death, any revascularization (target or nontarget vessel or lesion), and definite or probable stent thrombosis. Definitions of all endpoints have been previously published.¹¹⁻¹³ Briefly, cardiovascular death was defined as death resulting from cardiovascular causes. Periprocedural MI was defined as a creatine kinase-MB elevation $>3\times$ the upper limit of normal, or a troponin elevation $>21\times$ upper limit of normal if creatine kinase-MB was unavailable. MI during unplanned revascularization was classified as non-procedural. Target lesion or vessel referred to those receiving revascularization at the index procedure. All events were adjudicated by an independent committee blinded to treatment allocation.

An independent angiographic core laboratory performed quantitative coronary angiography, computed Synergy Between Percutaneous Coronary Intervention with Taxus and Cardiac Surgery (SYNTAX) scores, and performed offline QFR analysis for baseline and post-PCI angiograms in all patients. Based on these measurements, "pre-PCI QFR-concordant vessel selection" was defined as the treatment of all vessels with QFR ≤ 0.80 and the deferral of all vessels with QFR >0.80 .

Follow-up was originally scheduled at 1 month, 6 months, and annually at 1, 2, and 3 years. In November 2022, with additional funding, follow-up was extended to 5 years, with all centers participating to collect prespecified adverse event data at years 4 and 5. All patients were reconsented for this extended follow-up. The present report summarizes the final 5-year follow-up outcomes.

STATISTICAL ANALYSIS. Analyses were conducted in the intention-to-treat population. Continuous variables are presented as mean $\pm$ SD or median (IQR) and were compared using 2-sample *t*-tests or the Wilcoxon rank-sum test, as appropriate. Categorical variables are presented as counts and percentages and were compared using the likelihood ratio chi-square test or Fisher exact test. Event rates over the 5-year follow-up were estimated using Kaplan-Meier curves and compared using log-rank test. For cardiovascular-specific outcomes, including cardiovascular death, MI, and ID-Revasc, non-cardiovascular death was treated as a censoring event

FIGURE 1 Patient Flow Chart of FAVOR III China Trial



PCI = percutaneous coronary intervention; QFR = quantitative flow ratio.

at the time of death in the time-to-event analyses. Landmark analyses were performed to examine outcomes within the first 2 years and from 2 to 5 years, testing the interaction between treatment and time. HRs with 95% CIs were calculated using Cox proportional hazards models. Patients lost to follow-up were censored at the date of last contact. A sensitivity analysis excluding those lost to follow-up was performed to investigate the potential influence of loss to follow-up.

Prespecified subgroup analyses were conducted, and tests for interaction were performed for the following variables: age, sex, diabetes mellitus, presentation with acute coronary syndrome, smoking history, body mass index, left ventricular ejection fraction, lesion location, lesion length, RVD, stenosis severity, multivessel disease, calcified lesion,

bifurcation lesion, tandem lesion, angulated or tortuous lesion, QFR gray zone (0.75-0.85), QFR-based functional SYNTAX score, pre-PCI simulated residual QFR, QFR-based residual functional SYNTAX score, center experience with invasive physiology, and learning experience with QFR.

All statistical tests were 2-sided, and a $P < 0.05$ was considered statistically significant. Analyses were performed using SAS software, version 9.4 (SAS Institute).

RESULTS

PARTICIPANTS AND FOLLOW-UP. A total of 3,825 patients were enrolled from 26 centers in China between December 2018 and January 2020 (Figure 1). Of these, 1,913 patients were randomized to QFR-guided PCI and 1,912 to angiography-guided PCI.

TABLE 1 Baseline and Angiographic Characteristics of Patients Who Underwent Randomization

	QFR Guidance (n = 1,913)	Angiography Guidance (n = 1,912)	P Value
Demographics			
Age	62.7 ± 10.1	62.7 ± 10.2	0.81
Male	1,349 (70.5)	1,350 (70.6)	0.95
Risk factors			
Diabetes mellitus	648 (33.9)	647 (33.8)	0.98
Hypertension	1,270 (66.4)	1,252 (65.5)	0.55
Hypercholesterolemia	729 (38.1)	728 (38.1)	0.98
Current smoker	574 (30.0)	568 (29.7)	0.84
Family history of CAD	147 (7.7)	149 (7.8)	0.90
Previous MI	179 (9.4)	179 (9.4)	1.00
Previous PCI	485 (25.4)	466 (24.4)	0.48
Clinical presentation			
Unstable angina ^a	1,111 (58.1)	1,110 (58.1)	0.99
Post-MI (within 30 d)	102 (5.3)	105 (5.5)	0.83
Left ventricular ejection fraction, median (IQR), %	63.0 (61.0-66.0)	63.0 (60.0-66.0)	0.58
Angiographic findings			
Multivessel disease	1,023 (53.5)	1,043 (54.6)	0.51
Anatomic SYNTAX score	9.3 ± 6.0	9.6 ± 6.3	0.23
Functional SYNTAX score ^b	8.1 ± 6.3	8.0 ± 6.6	0.83
Offline QFR value (angiographic core laboratory assessment)	0.72 ± 0.15 (2,775 vessels)	0.72 ± 0.15 (2,789 vessels)	0.90
Procedures			
PCI performed	1,731 (90.5)	1,895 (99.1)	<0.0001
Number of stents placed per patient	1.45 ± 1.02	1.58 ± 0.97	<0.0001
Stent length, mm	42.7 ± 26.3	41.9 ± 26.3	0.36
Stent diameter, mm	3.03 ± 0.41	3.01 ± 0.41	0.34
PCI lesion success ^c	2,245/2,267 (99.0)	2,561/2,580 (99.3)	0.38
PCI procedure success ^d	1,657/1,731 (95.7)	1,796/1,895 (94.8)	0.18
Patients with intended vessel deferral or unintended vessel treatment	445 (23.3)	119 (6.2)	<0.0001
Deferral of at least 1 vessel originally intended for PCI	375 (19.6)	100 (5.2)	<0.0001
Treatment of at least 1 vessel not originally intended for PCI	85 (4.4)	28 (1.5)	<0.0001
Residual anatomic SYNTAX score	2.4 ± 3.6	2.4 ± 4.0	0.49
Residual functional SYNTAX score	0.7 ± 2.3	1.0 ± 2.8	<0.0001
Residual functional SYNTAX score of 0	1,669/1,894 (88.1)	1,552/1,887 (82.2)	<0.0001

Values are mean ± SD, median (IQR), n/N (%), or n (%). ^aReported by each site. ^bThe functional SYNTAX score was calculated by summing the segmental anatomic SYNTAX scores only in vessels with functional ischemia, defined as offline quantitative flow ratio ≤0.80 and assessed by the angiographic core laboratory. ^cDefined as residual stenosis <30% after stent implantation or <50% after balloon angioplasty, by visual estimation, with TIMI flow grade 3 in the treated vessel. ^dDefined as lesion success in all treated lesions without in-hospital MACE (up to 7 d).

CAD = coronary artery disease; MACE = major adverse cardiac events; MI = myocardial infarction; PCI = percutaneous coronary intervention; QFR = quantitative flow ratio; SYNTAX = Synergy between Percutaneous Coronary Intervention with Taxus and Cardiac Surgery; TIMI = Thrombolysis In Myocardial Infarction.

Baseline clinical and angiographic characteristics were balanced between groups (Table 1). The mean age was 62.7 years, and approximately 70% of patients were male. Cardiovascular risk factors were well matched, with 9.4% having prior MI and ~25% having prior PCI. In terms of clinical presentation, 58% had unstable angina and 5% presented with a recent MI (within 30 days). More than one-half of the patients had multivessel disease, with a mean SYNTAX score of 9.5 and a mean preprocedural QFR of 0.72, as assessed offline by the imaging core laboratory.

Fewer PCIs were performed in the QFR-guided group than in the angiography-guided group (90.5% vs 99.1%; $P < 0.0001$), and fewer stents were implanted (1.45 ± 1.02 vs 1.58 ± 0.97 ; $P < 0.0001$). Lesion and procedural success rates were comparable between groups. In the QFR-guided group, treatment strategy was modified in 445 (23.3%) patients: 375 (19.6%) patients had deferral of at least 1 vessel that was initially intended for PCI, and 85 (4.4%) underwent PCI in a vessel not originally planned for treatment. In the angiography-guided group, operator-driven strategy changes only occurred in

TABLE 2 Outcomes at 5-Year Follow-Up

	QFR Guidance (n = 1,913)	Angiography Guidance (n = 1,912)	HR (95% CI)	P Value
MACE	323 (17.5)	390 (21.1)	0.80 (0.69-0.92)	0.002
Death from any cause	91 (5.0)	75 (4.2)	1.20 (0.89-1.63)	0.24
MI	107 (5.8)	167 (9.0)	0.63 (0.49-0.80)	<0.0001
ID-Revasc	173 (9.6)	217 (12.0)	0.78 (0.64-0.95)	0.02
MACE excluding PMI	276 (15.0)	324 (17.6)	0.82 (0.70-0.97)	0.02
Cardiovascular death	53 (3.0)	41 (2.3)	1.28 (0.85-1.93)	0.23
Noncardiovascular death	38 (2.1)	34 (1.9)	1.11 (0.70-1.76)	0.67
Nonprocedural MI	52 (2.9)	90 (4.9)	0.57 (0.40-0.80)	0.001
Any revascularization	210 (11.6)	245 (13.5)	0.84 (0.70-1.01)	0.07
Target vessel revascularization	107 (5.9)	133 (7.4)	0.79 (0.62-1.02)	0.08
Ischemia-driven	90 (5.0)	120 (6.7)	0.74 (0.56-0.97)	0.03
Target lesion revascularization	76 (4.2)	103 (5.7)	0.73 (0.54-0.98)	0.04
Ischemia-driven	70 (3.9)	94 (5.2)	0.74 (0.54-1.00)	0.052
Nontarget lesion revascularization	37 (2.0)	34 (1.9)	1.08 (0.68-1.73)	0.74
Ischemia-driven	25 (1.4)	30 (1.7)	0.83 (0.49-1.41)	0.48
Nontarget vessel revascularization	114 (6.2)	123 (6.7)	0.92 (0.71-1.18)	0.50
Ischemia-driven	88 (4.8)	103 (5.7)	0.84 (0.63-1.12)	0.24
Stent thrombosis, definite or probable	11 (0.6)	22 (1.2)	0.50 (0.24-1.02)	0.06

Event rates are expressed as n (Kaplan-Meier estimated event rates). HRs represent the risks for the QFR-guided vs angiography-guided groups.
ID-Revasc = ischemia-driven revascularization; PMI = periprocedural myocardial infarction; other abbreviations as in [Table 1](#).

119 (6.2%) patients. The residual anatomical SYNTAX score was low and similar between groups (2.4 ± 3.6 vs 2.4 ± 4.0 ; $P = 0.49$). However, the residual functional SYNTAX score was significantly lower in the QFR-guided group (0.7 ± 2.3 vs 1.0 ± 2.8 ; $P < 0.0001$), with more patients achieving functional complete revascularization (defined as residual functional SYNTAX score = 0; 88.1% vs 82.2%; $P < 0.0001$).

Five-year follow-up was completed in 1,808 of 1,913 (94.5%) patients in the QFR-guided group and 1,798 of 1,912 (94.0%) patients in the angiography-guided group. Characteristics of patients with completed follow-up are shown in [Supplemental Table 1](#), demonstrating similar findings as in the overall study population.

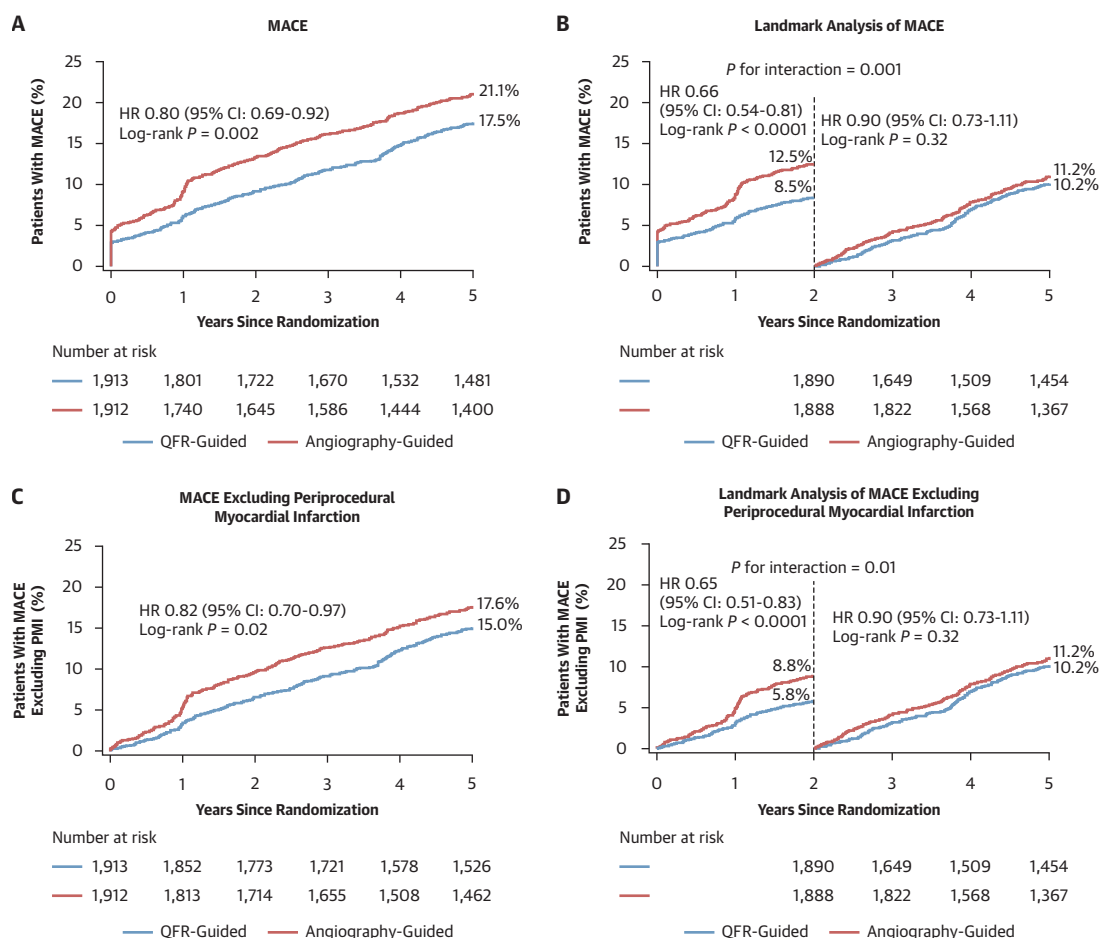
CLINICAL OUTCOMES. At 5 years, the incidence of MACE was significantly lower in the QFR-guided group than in the angiography-guided group (17.5% vs 21.1%; HR: 0.80; 95% CI: 0.69-0.92; $P = 0.002$), driven by reductions in MI (5.8% vs 9.0%; HR: 0.63; 95% CI: 0.49-0.80; $P < 0.0001$) and ID-Revasc (9.6% vs 12.0%; HR: 0.78; 95% CI: 0.64-0.95; $P = 0.02$). All-cause mortality rate did not differ between groups (5.0% vs 4.2%; HR: 1.20; 95% CI: 0.89-1.63; $P = 0.24$). The 5-year rate of MACE excluding periprocedural MI was also lower in the QFR-guided group (15.0% vs 17.6%; HR: 0.82; 95% CI: 0.70-0.97; $P = 0.02$) ([Table 2](#), [Figure 2](#)). Similar results were observed in patients completing 5-year follow-up ([Supplemental Table 2](#)).

Landmark analyses showed that the benefit of QFR guidance was concentrated in the first 2 years (MACE: HR: 0.66; 95% CI: 0.54-0.81; $P < 0.0001$), whereas outcomes between 2 and 5 years were comparable between the groups (10.2% vs 11.2%; HR: 0.90; 95% CI: 0.73-1.11; $P = 0.32$; $P_{\text{interaction}} = 0.001$) ([Figure 2](#)). Similar patterns were observed for MACE excluding periprocedural MI ($P_{\text{interaction}} = 0.01$) ([Figure 2](#)), MI ($P_{\text{interaction}} = 0.007$), and ID-Revasc ($P_{\text{interaction}} = 0.006$) ([Supplemental Figure 1](#)).

SUBGROUP ANALYSIS. Among patients who underwent PCI, QFR guidance was associated with a lower risk of MACE compared with angiographic guidance (18.3% vs 21.2%; HR: 0.83; 95% CI: 0.72-0.97; $P = 0.02$). No significant difference was observed among patients who were deferred from PCI (9.3% vs 6.7%; HR: 1.29; 95% CI: 0.17-9.75; $P = 0.80$). Furthermore, QFR guidance markedly reduced MACE in patients with treatment strategy changes between intended and actual treatment (16.2% vs 33.5%; HR: 0.42; 95% CI: 0.28-0.62; $P < 0.0001$), whereas a borderline benefit was observed in those without strategy changes (17.9% vs 20.3%; HR: 0.85; 95% CI: 0.72-0.999; $P = 0.049$) ([Table 3](#)).

QFR guidance consistently reduced MACE risk across most prespecified subgroups. Significant interactions were present suggesting less benefit of QFR guidance in patients who never smoked and in bifurcation lesions ([Figure 3](#)).

FIGURE 2 5-Year Outcomes for the Primary and Major Secondary Endpoint



Cumulative incidence curves are shown for MACE (A) and MACE excluding PMI (C), along with landmark analyses for MACE (B) and MACE excluding PMI (D). HRs represent the risks for QFR-guided vs angiography-guided PCI. MACE was defined as a composite of all-cause death, MI, or ID-Revasc. ID-Revasc = ischemia-driven revascularization; MACE = major adverse cardiac events; MI = myocardial infarction; PMI = periprocedural myocardial infarction; other abbreviations as in [Figure 1](#).

DISCUSSION

In the randomized, sham-controlled, multicenter FAVOR III China trial, a QFR-guided PCI strategy reduced the 5-year incidence of MACE by 20% compared with angiography-guided PCI, mainly driven by fewer MIs and IDRs, whereas long-term rates of all-cause mortality were similar in both groups. Landmark analysis indicated that the clinical benefit predominantly accrued within the first 2 years and that event rates were similar between 2 and 5 years. These findings provide the longest-term evidence to date supporting late benefits of QFR-guided PCI and confirm the safety of QFR-based lesion deferral ([Central Illustration](#)).

Our results extend and complement prior evidence establishing the value of physiology-guided revascularization with wire-based indices (FFR, instantaneous wave-free ratio, or resting full-cycle ratio) relative to angiographic guidance. Importantly, FAVOR III China demonstrates that PCI guidance with an angiography-derived index (QFR) that can be derived rapidly and without a pressure wire can improve lesion selection and long-term clinical outcomes compared with angiography guidance alone.^{12,13} FAVOR III China was a large, randomized, multicenter study with a sham-controlled design, high follow-up completeness, and independent core laboratory assessments, ensuring methodologic rigor and supporting the robustness of its conclusions.

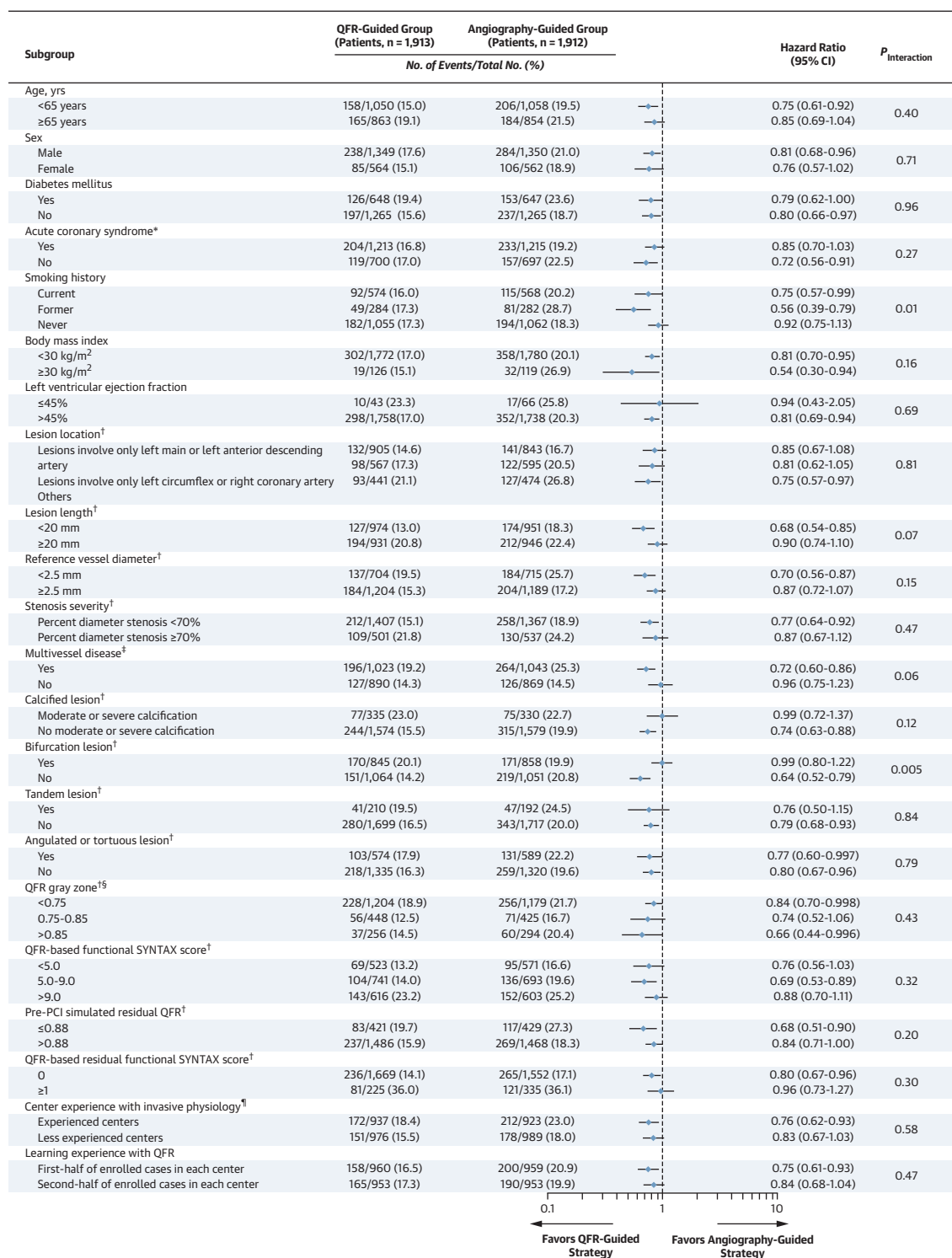
TABLE 3 5-Year Outcomes According to PCI Deferral and Strategy Change Status

	QFR Guidance (n = 1,913)	Angiography Guidance (n = 1,912)	HR (95% CI)	P Value
Patients with PCI performed	N = 1,731	N = 1,895		
MACE	307 (18.3)	389 (21.2)	0.83 (0.72-0.97)	0.02
Death from any cause	87 (5.3)	75 (4.2)	1.26 (0.92-1.71)	0.15
MI	102 (6.1)	167 (9.0)	0.66 (0.52-0.84)	0.001
ID-Revasc	162 (9.9)	216 (12.0)	0.81 (0.66-0.99)	0.04
MACE excluding PMI	260 (15.6)	323 (17.7)	0.85 (0.72-1.003)	0.054
Patients with PCI deferral	N = 182	N = 17		
MACE	16 (9.3)	1 (6.7)	1.29 (0.17-9.75)	0.80
Death from any cause	4 (2.3)	0 (0)	-	0.71
MI	5 (3.0)	0 (0)	-	0.67
ID-Revasc	11 (6.4)	1 (6.7)	0.89 (0.12-6.90)	0.91
MACE excluding PMI	16 (9.3)	1 (6.7)	1.29 (0.17-9.75)	0.80
Patients with strategy change	N = 445	N = 119		
MACE	70 (16.2)	38 (33.5)	0.42 (0.28-0.62)	<0.0001
Death from any cause	19 (4.5)	4 (3.7)	1.24 (0.42-3.65)	0.70
MI	26 (6.1)	17 (14.9)	0.38 (0.21-0.70)	0.002
ID-Revasc	42 (9.9)	24 (21.4)	0.42 (0.26-0.70)	0.001
MACE excluding PMI	61 (14.2)	30 (26.7)	0.48 (0.31-0.75)	0.001
Patients without strategy change	N = 1,468	N = 1793		
MACE	253 (17.9)	352 (20.3)	0.85 (0.72-0.999)	0.049
Death from any cause	72 (5.2)	71 (4.2)	1.23 (0.89-1.71)	0.22
MI	81 (5.7)	150 (8.6)	0.65 (0.50-0.85)	0.002
ID-Revasc	131 (9.4)	193 (11.4)	0.82 (0.65-1.02)	0.07
MACE excluding PMI	215 (15.3)	294 (17.1)	0.87 (0.73-1.03)	0.11

Event rates are expressed as n (Kaplan-Meier estimated event rates). HRs represent the risks for the QFR-guided vs angiography-guided groups.
Abbreviations as in [Tables 1 and 2](#).

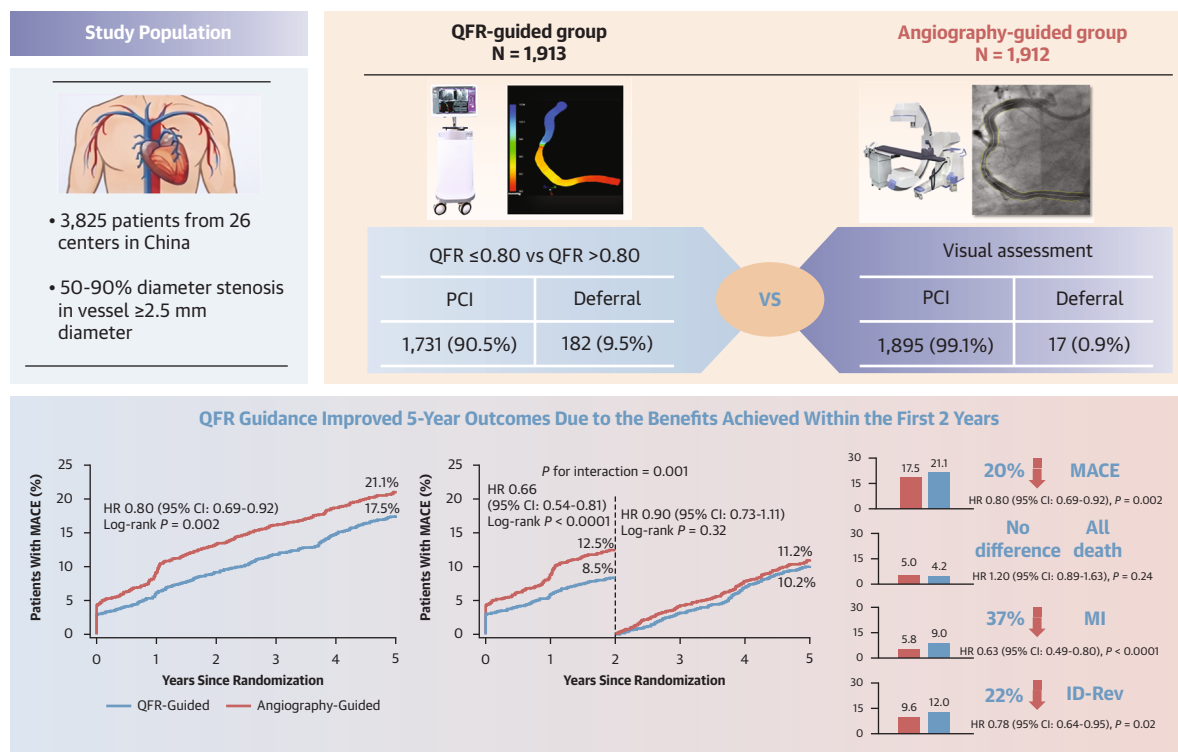
Mechanistically, the principal advantage of QFR guidance appears to be optimization of the index revascularization strategy: QFR avoided unnecessary stenting (fewer stents and equipment costs), thereby reducing vascular injury and procedure-related MI, and diminished residual functional ischemia, all plausible mediators of the observed early reductions in MI and repeat revascularization during early follow-up.^{16,17} A prespecified and methodologic strength of FAVOR III China was the prospective declaration of operator-intended target lesions before randomization; this permitted quantification of how QFR altered operator behavior. Patients in whom treatment plans were revised after QFR assessment derived particularly large gains, emphasizing that improved precision in target-vessel selection (ie, achieving QFR-concordant treatment) is a key determinant of the favorable prognosis associated with QFR guidance. Offline core-laboratory analyses corroborated that QFR-concordant revascularization (treating QFR ≤ 0.80 lesions and deferring QFR > 0.80 lesions) was associated with substantially lower event rates than QFR-nonconcordant strategies (2-year MACE: 8.8% vs 17.2% in concordant vs nonconcordant groups).¹³

Landmark analysis from the present study demonstrates that the divergence in clinical outcomes occurred predominantly within the first 2 years, consistent with prior 2-year analyses showing the incremental benefits during year 0-1 and year 1-2.¹³ However, no further benefits of QFR guidance (nor loss of this early benefit) were observed between years 2 and 5 after randomization. Two complementary interpretations are plausible. Physiology-guided lesion selection may primarily mitigate early, procedure-related and early spontaneous ischemic events through more precise revascularization; beyond this period, outcomes appear to be determined largely by the natural history of CAD and comorbidities rather than the index PCI strategy. Alternatively, crossover and subsequent revascularizations in the angiography-guided arm may attenuate differences over time. The impact of cross-overs on diminishing the early benefits of physiological guidance have also been observed in FAME II and other trials.^{18,19} Additional insight into the clustering of events between 12 and 15 months is provided in [Supplemental Table 3](#). Although routine angiographic follow-up was not protocol mandated, the open-label design may still have influenced downstream referral patterns.

FIGURE 3 Subgroup Analyses for the Primary Endpoint MACE at 5 Years

Subgroup analyses are exploratory, and *P* values were not adjusted for multiplicity. Data are presented as n/N (%). The HR (with 95% CI) for MACE in the QFR-guided group vs the angiography-guided group is shown in prespecified subgroups. *Includes unstable angina or MI within 30 days of the procedure. [†]Assessed by the angiographic core laboratory. [‡]Reported by site investigators. [§]Participants were categorized in the QFR 0.75-0.85 subgroup if QFR was 0.75-0.85 for all assessed vessels; otherwise, if participants had at least 1 vessel with QFR <0.75, they were categorized in the QFR <0.75 subgroup. [¶]Experienced center was defined as more than 100 annual invasive pressure wire cases. SYNTAX = Synergy between Percutaneous Coronary Intervention with Taxus and Cardiac Surgery; other abbreviations as in [Figures 1 and 2](#).

CENTRAL ILLUSTRATION 5-Year Follow-Up From the FAVOR III China Randomized Trial



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Among 3,825 patients enrolled at 26 centers in China with 50% to 90% diameter stenosis in vessels ≥ 2.5 mm, a QFR-guided strategy was associated with a significantly lower incidence of MACE than angiography guidance through 5 years. The benefit of QFR guidance was mainly observed within the first 2 years, with significant reductions in MI and ID-Revasc. ID-Revasc = ischemia-driven revascularization; MACE = major adverse cardiac events; MI = myocardial infarction; PCI = percutaneous coronary intervention; QFR = quantitative flow ratio.

Nonetheless, the excess events around 1 year in the angiography-guided group were more likely clinically driven, potentially reflecting greater residual ischemia or recurrent symptoms after less precise index revascularization rather than routine surveillance alone. Importantly, all ischemia-driven revascularization events underwent physiological assessment and quantitative coronary angiography review, reducing the likelihood that the observed difference was solely surveillance-related.

The present results must be interpreted in the context of the FAVOR III Europe trial in which FFR guidance was superior to QFR guidance to reduce 1-year MACE after PCI.²⁰ Although follow-up data beyond 1 year from this study have not been reported, several factors may explain why QFR guidance was strongly positive in FAVOR III China (compared with angiography guidance) but not in

FAVOR III Europe (compared with FFR guidance). First, these trials addressed different clinical implementation scenarios. FAVOR III China evaluated the impact of introducing an angiography-derived physiological strategy in routine angiography-guided practice, whereas FAVOR III Europe compared 2 physiology-guided approaches in a setting with systematic functional assessment. Differences in comparators, patient and lesion characteristics, and workflow constraints across trials likely account for variability in observed effect sizes. Second, the overall physiological severity of the FAVOR III China population should be considered. The mean pre-PCI QFR was 0.72, similar to the mean FFR of 0.71 in FAME,¹⁴ consistent with a population with relatively severe physiologically significant disease. Exploratory analyses from FAVOR III China indicated attenuated benefit in physiologically intermediate

lesions (the “gray zone,” eg, QFR 0.75-0.85).²¹ This finding implies that a hybrid strategy may be appropriate in clinical practice, for example, QFR could be used for first-line functional assessment, with invasive FFR reserved for lesions within the QFR gray zone. Importantly, QFR and FFR provide physiological assessment, whereas intravascular imaging offers complementary information for lesion characterization and procedural optimization, particularly in complex anatomy. Third, technical aspects, including QFR software generation, angiographic acquisition quality, and operator experience, may have influenced diagnostic accuracy and subsequent outcomes.²² Ongoing advances in software, such as Murray law-based QFR (μ FR), artificial intelligence-assisted single-view reconstruction, and workflow integration, may further narrow the gap and define the optimal role of QFR in global practice.^{23,24} Taken together, these trials should be viewed as complementary: FAVOR III China shows that QFR represents a substantial advance over angiography-guidance alone in routine practice (leading to improved long-term results), whereas FAVOR III Europe underscores that invasive FFR remains the physiological gold standard in physiologically borderline lesions.

Moreover, use of QFR as the initial guidance modality offers several important clinical advantages. As a pragmatic, wire-free tool that lowers logistical and financial barriers to physiology-guided PCI, QFR offers the potential to increase uptake of physiology assessment in everyday practice, especially where invasive pressure wire use is limited by cost, time, potential side effects, or expertise.^{25,26} The sustained benefit of QFR guidance in reducing 5-year MACE after PCI demonstrated in the present large, multicenter trial consolidates QFR as a practical, safe, and scalable tool for physiology-based revascularization in daily practice.^{1,27} Future research priorities include: 1) prospective evaluation of hybrid algorithms (first-line QFR with selective invasive FFR for gray-zone lesions); 2) head-to-head assessments of next-generation angiography-derived indices (eg, μ FR) vs invasive physiology in diverse populations; 3) integration of lesion biomechanics (such as radial wall strain) and intracoronary imaging to identify “vulnerable” but non-flow-limiting plaques that may underlie some late MACE events;²⁸⁻³⁰ and 4) formal cost-effectiveness and implementation studies to define the optimal pathways for widespread adoption.³¹

STUDY LIMITATIONS. First, the study population consisted of patients with stable angina and stabilized acute coronary syndromes with mostly low SYNTAX scores; patients with cardiogenic shock and highly complex anatomy were excluded, and extrapolation to these higher-risk cohorts should be made with caution. Second, the trial was originally designed with 3-year follow-up; extension to 5 years was investigator-initiated, and information on medication use during the extended period was not systematically collected. Third, the trial used a first-generation QFR algorithm that required 2 angiographic projections per vessel; newer single-view μ FR and artificial intelligence-enhanced reconstructions may further improve workflow and accuracy³² but were not evaluated here. Fourth, PCI operators were aware of group assignments, which may have introduced procedural bias. Nonetheless, prerandomization declaration of the intended treatment strategy and the sham-controlled design ensured robust blinding of patients and clinical assessors during follow-up, thereby minimizing placebo, Hawthorne, and ascertainment biases. Fifth, the limited use of intracoronary imaging precludes insight into how imaging guidance might have affected the outcomes of a QFR-guided strategy. Sixth, the results may not be generalizable to excluded cohorts, such as patients with moderate and severe chronic kidney disease or severe vessel tortuosity. Seventh, subgroup analyses were not adjusted for multiplicity, and thus the findings of possibly lower benefit from QFR guidance in never smokers and bifurcation lesions should be interpreted cautiously. Finally, the angiography-guided strategy relied on operator visual assessment and local practice patterns, which are inherently subjective and may limit the generalizability of the findings to other countries or different clinical practice environments.

CONCLUSIONS

The 5-year results from the FAVOR III China trial demonstrate that QFR-guided PCI yields durable reductions in MACE, MI, and the need for repeat revascularization through 5 years, without compromising safety. These findings support QFR as a practical and scalable first-line approach to physiology-guided revascularization, particularly in settings where invasive pressure wire assessment is constrained. Future studies should focus on

integrating next-generation angiography-derived indices with invasive physiology and formal cost-effectiveness analyses to enable personalized revascularization pathways in contemporary practice.

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PERSPECTIVES

COMPETENCY IN PATIENT CARE AND PROCEDURAL

SKILLS: In patients with angiographically intermediate CAD undergoing PCI, a QFR-guided strategy improves long-term clinical outcomes compared with angiography-guided decision making, with reductions in MI and ID-Revasc over 5-year follow-up.

TRANSLATIONAL OUTLOOK: Prospective studies should define the role of hybrid physiological strategies that combine angiography-derived indices with selective invasive assessment to enhance diagnostic precision while preserving procedural efficiency, and incorporate formal health-economic evaluations to support personalized revascularization in real-world practice.

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- KEY WORDS** angiography, coronary artery disease, long-term outcomes, percutaneous coronary intervention, quantitative flow ratio
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- APPENDIX** For supplemental tables and figures, please see the online version of this paper.